

EEI4369 Mobile App User Manual

Build buddy

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1. Introduction and Problem Statement

1.1 Brief introduction to the app.

Build Buddy is a mobile app that connects contractors and homeowners with nearby skilled and unskilled laborers on demand. Using GPS, skill filtering, and secure in-app payments, it makes hiring, tracking, and communicating with workers fast, easy, and reliable—all from your phone

1.2 Overview of the real-world problem the app addresses.

Build Buddy addresses these problems by providing a mobile platform that:

- Connects clients with nearby skilled and unskilled laborers on demand using GPS and skill filtering.
- Enables secure, in-app communication and multimedia sharing for clear job instructions and updates.
- Streamlines booking, scheduling, and payment processes, ensuring accountability and reducing misunderstandings.
- Offers ratings, reviews, and verified profiles to build trust and transparency between parties.
- By digitizing and centralizing the hiring and management process, Build Buddy reduces the risks of poor-quality work, payment disputes, and project delays, making construction projects more efficient and reliable for everyone involved.

1.3 Target users and the context in which the app will be used.

Build Buddy primarily targets two main user groups:

- **Clients:** This includes homeowners who want to manage and build their own homes without hiring traditional builders, as well as contractors who need skilled and unskilled laborers on demand. These users seek greater control, cost savings, and transparency in their construction projects.
- **Laborers:** Skilled tradespeople such as electricians, plumbers, carpenters, and welders, as well as unskilled helpers and loaders, who are available for hire through the app's platform.

Context of Use

- The app is designed for use in real-world construction and home-building environments where clients need to quickly find, hire, and manage laborers nearby.
- It is used on mobile devices, leveraging GPS for location-based matching and live tracking of workers en route to jobs.
- Users interact with the app to communicate securely via chat and voice calls, share multimedia (images, videos, voice notes) to clarify job details, and manage bookings, scheduling, and payments all within the platform.
- Build Buddy is especially suited for owner-builders—people without formal construction experience who want to take charge of their home building projects with expert guidance and step-by-step workflows.
- The app also appeals to budget-conscious homeowners, experienced DIYers, and investors looking to save on builder margins and gain direct access to verified trades and suppliers.
- By providing a comprehensive, easy-to-use platform that integrates project management, communication, and payment, Build Buddy fits into the daily workflow of managing construction projects from planning through completion, empowering users to build efficiently and confidently

2. Proposed Solution / Your App Overview

2.1 Description of the mobile app solution.

Build Buddy is a mobile app that connects homeowners and contractors with skilled and unskilled laborers nearby using GPS-based matching and skill filters. It offers secure in-app communication, instant booking, scheduling, and payment features, along with ratings and reviews to build trust. The app streamlines the hiring and management of construction labor, making the process faster, more transparent, and efficient—all from your phone.

2.2 Key features and functionalities

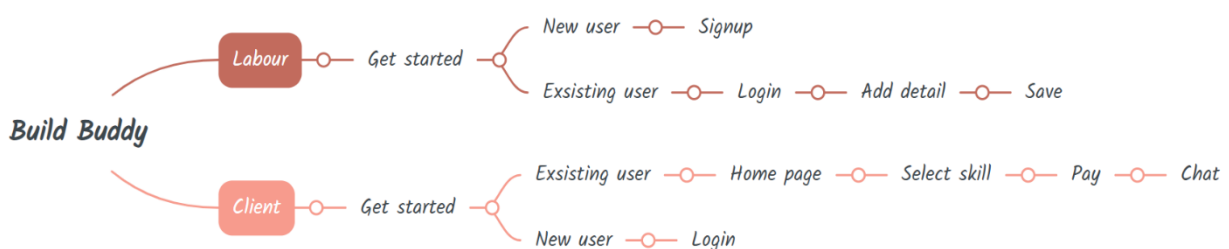
- **User Roles:** Supports both clients (contractors, homeowners) and laborers (skilled and unskilled) for flexible project needs.
- **GPS-Based Matching:** Locates nearby laborers using Google Maps API, with live tracking of workers en route to job sites.
- **Skill-Based Search:** Filters workers by skill type, experience level, and availability for precise hiring.
- **Communication:** Enables chat and voice calls, along with multimedia sharing (images, videos, voice notes) for clear job instructions and updates.
- **Booking & Scheduling:** Allows instant booking or scheduling for later, with auto-reminders and calendar integration to keep projects on track.
- **Secure Payments:** Processes payments through Stripe, local bank APIs, or mobile wallets, with automatic commission deduction for transparency.
- **Ratings and Reviews:** Lets clients and laborers rate each other, building trust and accountability within the platform.
- **Profile and Database Management:** Stores user profiles, chat histories, shared documents, and payment details securely for easy access and record-keeping

2.3 Goals and objectives of the app.

- Empower Homeowners and Contractors: Enable anyone, regardless of experience, to manage and build their own home or project by providing step-by-step guidance and access to essential resources.
- Streamline Labor Hiring: Make it fast and easy to find, hire, and manage skilled and unskilled laborers nearby using GPS and skill-based matching.
- Increase Transparency and Trust: Foster a transparent construction process through real-time updates, ratings, reviews, and verified profiles, building trust among all parties involved.
- Enhance Communication: Provide tools for clear and efficient communication, including chat, voice calls, and multimedia sharing, to reduce misunderstandings and improve project coordination.
- Simplify Project Management: Offer intuitive workflows, scheduling, and task management features that guide users through each stage of the construction process.
- Reduce Costs: Help users save up to 20% or more by eliminating builder margins and offering access to wholesale-priced materials and services.
- Support Collaboration: Create a collaborative environment where homeowners, contractors, and suppliers can work together seamlessly, sharing information and updates in real time.
- Deliver Expert Support: Ensure users have access to professional advice and on-demand guidance throughout their building journey, boosting confidence and quality outcomes

3. Mind Map of App Functions

3.1 Visual mind map showing the structure and flow of the app



3.2 Overview of - Main Modules/ Sub-functions/ Navigation logic

Main Modules

User Roles

- Clients (Contractors, Homeowners)
- Laborers (Skilled and Unskilled)

Authentication

- Sign In
- Sign Up
- Profile Creation and Editing

Labor Search and Matching

- GPS-based location and live tracking
- Skill-based filtering (type, experience, availability)

Communication

- Chat and voice calls
- Multimedia sharing (images, videos, voice notes)

Booking & Scheduling

- Instant booking or schedule for later
- Auto-reminders and calendar integration

Payments

- In-app payments (Stripe, bank APIs, mobile wallets)
-

Ratings and Reviews

- Mutual rating system for clients and laborers

Database Integration

- Profile management
- Chat and messaging history
- Shared files and documents
- Payment and transaction records

Sub-functions

- Profile Management: Store and update user details, skills, experience, and payment preferences.
- Laborer Listing: Display available laborers with filters for skill, location, and availability.
- Job Posting: Clients can post job requirements with detailed descriptions and media.
- Laborer Selection: Clients view profiles, ratings, and select laborers.
- Job Tracking: Live updates and status tracking for ongoing jobs.
- Notifications: Auto-reminders for bookings, payment confirmations, and job status updates.

Navigation Logic

- Onboarding: Users choose their role (Client or Laborer) and complete sign-up or sign-in.
- Home Screen:
 - Clients: Access labor search, post jobs, view bookings, and manage payments.
 - Laborers: View job requests, accept/reject jobs, and update availability.
- Labor Search: Clients filter and select laborers, view profiles, and initiate bookings.
- Booking Flow: Confirm job details, make payments, and receive confirmation.
- Communication: Direct chat and call options become available after booking.
- Job Management: Both parties track job progress, share updates, and complete tasks.
- Ratings & Reviews: Prompted after job completion for both clients and laborers.

4. Implementation Details

- **Technologies Used:** Front-end (Android Studio)/Back-end (e.g.Firebase)
- **Database Used:** Firebase Firestore
- **APIs & Integrations:** Google Maps API

5. GitHub Repository

5.1 Link to the source code : [Munasinghe33/Android-final-project: Android final project](https://github.com/Munasinghe33/Android-final-project)

5.2 Repository structure :

- /app: Android app source code
- .idea
- gradle

5.3 Setup instructions

- Clone the repository.
- Open in Android Studio.
- Maps API keys.
- Run the app on an emulator or device.

6. How to Use the App (User Guide)

6.1 Starting screen

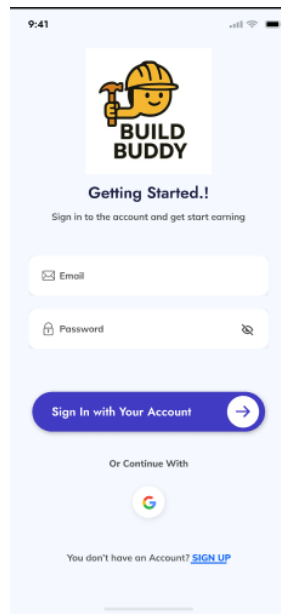
Screenshot:



- Description: Clean welcome screen with login/sign-up options.
- Business Flow: Directs new users to registration, returning users to login.

6.2 Login Page

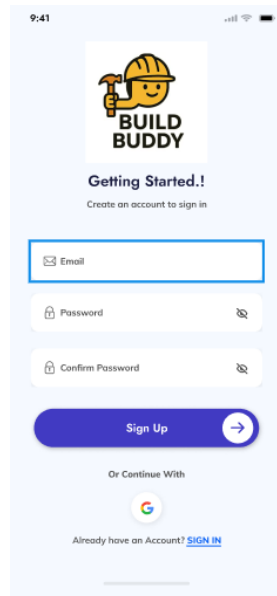
Screenshot



- Description: Email and password fields, 'Forgot Password' , 'Sign Up' link.
- Business Flow: Authenticates user and opens dashboard.

6.3 Sign Up Page

Screenshot:



- Description: Fields for email, password, and retype password.
- Business Flow: Registers user and redirects to Login.as a register User

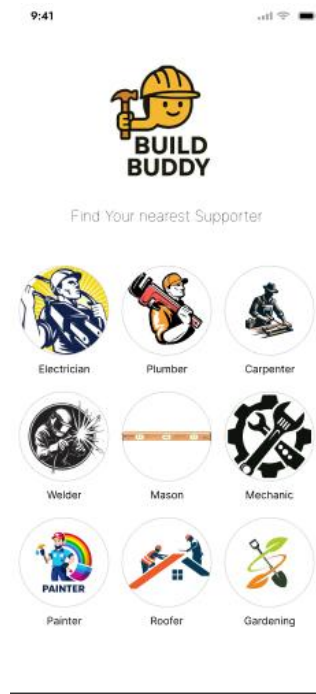
6.4 Forgot Password screen

Screenshot:

- Description: Fields for email, New password, and retype password, Routes to sign up and login.
- Business Flow: Reset password add new password then routs login.

6.5 Dashboard

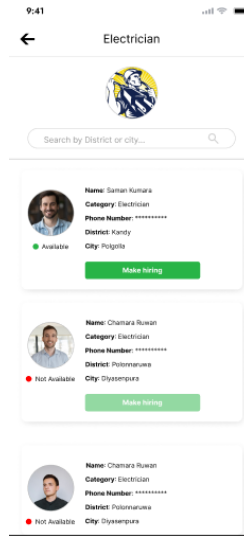
Screenshot:



- Description: Easily find and hire the nearest skilled worker for any home service with just a tap.
- Business Flow: Central hub for navigation.

6.6 Features 01 -Skill labours

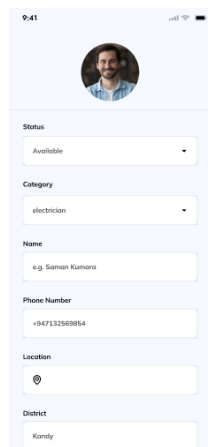
Screen shot



- Description: Indicate the skill type and Their details
- Business Flow- show skill labors

6.7 Features 2 – Add skill labor by their own experience

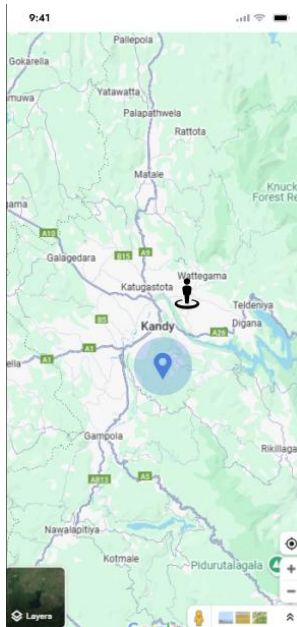
Screen shot



- Description: Labors add their skill

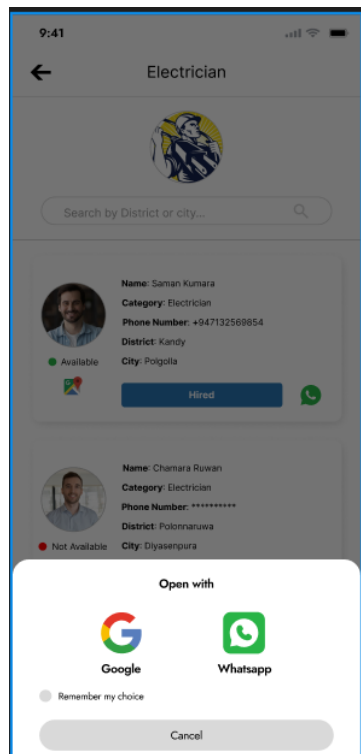
- Business Flow: show labor for add their skill

6.8 Google map Integration



- Description - client can search by location
- Business flow - help client to find labours

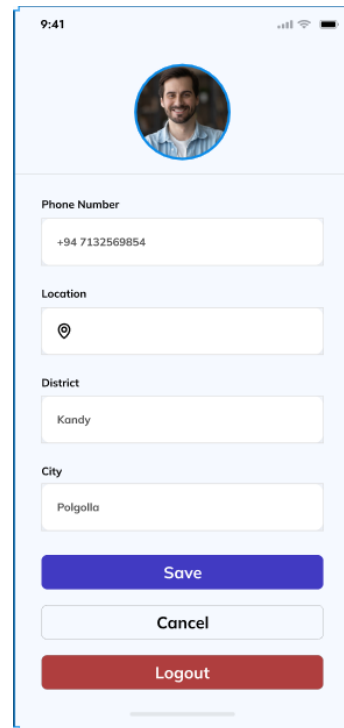
6.9 Client chat with labors



- Description - client can chat with labours
- Business flow – Can talk to labours and deal with them

6.10 log out

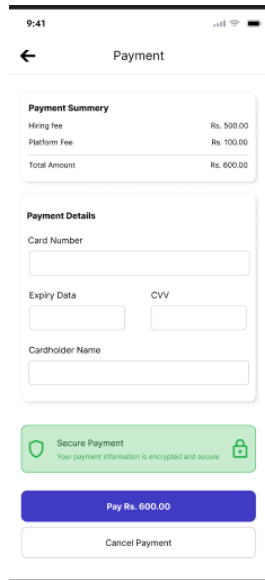
- Description: Log out button
- Business Flow: Ends session and returns to login.



A mobile application profile screen. At the top, the status bar shows the time 9:41, signal strength, Wi-Fi, and battery. Below the status bar is a circular profile picture of a man with a beard. Underneath the profile picture are four text input fields: 'Phone Number' with the value '+94 7132569854', 'Location' with a location pin icon, 'District' with the value 'Kandy', and 'City' with the value 'Polgolla'. At the bottom of the form are three buttons: a blue 'Save' button, a light blue 'Cancel' button, and a red 'Logout' button.

6.11 client can pay for site and get contact labor

Screen shot



The screenshot shows a mobile application interface for a payment screen. At the top, the status bar displays the time 9:41 and signal strength. The title bar shows a back arrow and the word 'Payment'. Below the title bar, there is a 'Payment Summary' section with a table listing 'Hiring fee' (Rs. 500.00), 'Platform Fee' (Rs. 100.00), and 'Total Amount' (Rs. 600.00). This is followed by a 'Payment Details' section with input fields for 'Card Number', 'Expiry Date', 'CVV', and 'Cardholder Name'. Below these fields is a green box with a shield icon and the text 'Secure Payment' and 'Your payment information is encrypted and secure'. At the bottom, there are two buttons: a blue 'Pay Rs. 600.00' button and a white 'Cancel Payment' button.

Payment Summary	
Hiring fee	Rs. 500.00
Platform Fee	Rs. 100.00
Total Amount	Rs. 600.00

Payment Details

Card Number

Expiry Date CVV

Cardholder Name

Secure Payment
Your payment information is encrypted and secure

Pay Rs. 600.00

Cancel Payment

- Description :Site can get income from payment
- Business Flow :Earn income

7. Testing and Results

7.1 Types of Testing Performed

- Unit Testing

Description:

Individual components such as user registration, laborer search, booking, communication, and payment processing were tested in isolation to ensure each function works as intended.

Outcome:

All core logic functions (e.g., user sign-up, laborer matching, job booking, and secure payment) passed unit tests with over 95% code coverage, confirming robust and reliable app behavior

Integration testing

Description:

Tested interactions between modules, such as syncing job bookings with the calendar and integrating Google Maps for locating nearby laborers and providing directions.

Outcome:

All major workflows (e.g., searching for and booking a laborer, navigating to the job site, and processing payments) performed as expected across tested devices, ensuring seamless integration and reliable user experience.

7.2 Issues

- **Internet Dependency:** The app requires a stable internet connection; poor connectivity can limit usage and cause delays.
- **High Data Usage:** Multimedia features (like images and videos) consume significant data, which may be challenging for users with limited plans or slow networks.
- **Verification and Security:** There is a risk of fake profiles or misuse, and sensitive data could be exposed if not properly protected.
- **Dispute Resolution:** Resolving conflicts between clients and laborers can be difficult through the app alone.
- **Tech Barriers:** Some laborers may lack smartphones or the technical skills needed to use the app effectively.
- **Device Limitations:** Advanced features may not work on older or less capable devices.

- **Commission Concerns:** The 10% commission deducted from payments may discourage some laborers from joining the platform.

8. Conclusion

8.1 Summary of the App's Value

Build Buddy is a comprehensive and user-friendly solution that streamlines the process of finding, hiring, and managing skilled and unskilled labor for construction projects. By offering features like skill-based search, GPS matching, secure payments, and transparent reviews, the app reduces the hassle and inefficiencies of traditional labor hiring, making construction more accessible and efficient for homeowners and contractors.

8.2 Future Improvements

- **Enhanced Verification:** Implement stronger identity checks and fraud prevention measures to improve trust and safety.
- **Offline Functionality:** Develop features that allow limited use without constant internet access.
- **Expanded Support:** Broaden device compatibility and provide training resources for less tech-savvy users.
- **Advanced Dispute Resolution:** Introduce better mechanisms for handling conflicts between users.
- **Localization:** Offer the app in multiple languages to reach a broader audience.
- **Flexible Commission Models:** Explore alternative fee structures to attract more laborers.

9 Appendix

- Figma Design Link: [Untitled – Figma](#)
- Git HUB Link : [Munasinghe33/Android-final-project: Android final project](#)